Lab 3: Component Modelling & Architectural Pattern Selection

Hospital Bed & ICU Allocation Dashboard

Name: Sharanabasava
SRN: PES2UG24CS457
Section: H
Subject: Software Engineering

Architecture Selection

We chose Client-Server Architecture for the Hospital Bed & ICU Allocation Dashboard.

Architectural Choice

The Hospital Bed & ICU Allocation Dashboard displays real-time ward occupancy, prioritizes ICU transfers using triage scores, and supports ambulance transfer workflows. The main users are the Triage Nurse and Hospital Administrator. In a Client-Server architecture, these users can access the system through client interfaces while the server handles the main hospital management functions.

Reason 1 – Centralized Bed Management

The system manages bed information such as Occupied, Cleaning, and Available. A central server can maintain the latest bed status and send the updated information to connected dashboard clients. This helps users see current bed availability.

Reason 2 – Triage and ICU Allocation

The system calculates triage severity scores from patient intake vitals and uses them to order the ER bed allocation waitlist. Centralized processing allows the server to perform this calculation and update the allocation information for the users.

Security Advantage

The server can control access to sensitive hospital information. Authentication and authorization can be used so that only authorized users can access patient, bed, ICU, and transfer information according to their roles.

Performance Benefit

Client-Server architecture supports fast communication between dashboard clients and the central system. WebSockets can be used to send bed-status changes to connected clients without requiring repeated page refreshes. The problem statement specifies that these updates should reach connected ER dashboard clients in under one second.

Conclusion

Client-Server Architecture provides a simple structure for the Hospital Bed & ICU Allocation Dashboard. It supports centralized bed management, real-time updates, triage-based prioritization, ICU allocation, and ambulance transfer coordination.

Design component diagrams:

